Shourov Joarder

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Education

The University of Texas at Austin (UT Austin) USA
Ph.D. in Electrical and Computer Engineering (ECE) *Aug 2026 – Present*
Advisor: Prof. Sandeep P. Chinchali

Bangladesh University of Engineering & Technology (BUET) Bangladesh
B.Sc. in Electrical and Electronic Engineering *Feb 2020 – March 2025*
CGPA: **3.88/4.00**

Research Interests

Multimodal LLM | VLMs | Reinforcement Learning | World Models

Publications

1. **Shourov Joader**, D. Sikdar, A. Akash, B. Battarai, P.K. Gyawali, *"Two is better than one: A Collapse-free Multi-Reward RLIF Training Framework ,"* — **NeurIPS 2026** *(arXiv)*.
2. **Shourov Joader**, T. Talukder, K. Hasan, *"Multi-Stage Residual-Aware Unsupervised Deep Learning Framework for Consistent Ultrasound Strain Elastography,"* *(arXiv)*.
3. S. Hasan, **Shourov Joader**, A. Nayem, H. Hasan, and S. A. Fattah, *"Multilingual Voice-Controlled Smart Wheelchair with Advanced Features," Published at* — *IEEE ECCE*. (DOI: *10.1109/ECCE64574.2025.11013785*)
4. A. Dhar, D. Sikder, A. Shovon, and **Shourov Joarder**, *"Skin Cancer Semantic Segmentation," Published at* — *IEEE ECCE*. (DOI: *10.1109/ECCE64574.2025.11013785*).

Research Experiences

GRA, UT Austin
Aug 2026 - Present
Supervisor : Prof. Sandeep P. Chinchali

Multimodal Theory of Mind (ToM) for Multi-Agent Collaboration
We are working to build an End-to-End MLLM-ToM that can infer about the mental-state (goal, belief) of an agent by looking at it's partial action traces.

- Aiming CVPR 2027 with this paper.

Research Collaborator, West Virginia University (WVU)
Mar 2026 - May 2026
Supervisor : Prof. Prashnna K. Gyawali

Multi-reward Unsupervised RLIF for LLM Reasoning (Preprint)
Designed a multi-reward RL framework using intrinsic reward signals to guide diverse reasoning paths in LLMs, mitigating mode collapse and improving self-correction with limited supervision. Our proposed RLIF training method scales training for longer steps.

- Proposed complementary intrinsic rewards to prevent reward hacking during on-policy training. We validated on multiple reasoning benchmarks.
- Paper at NeurIPS'26.

Undergraduate Thesis Student, EEE, BUET
Mar 2024 - Mar 2025
Supervisor : Dr. Kamrul Hasan

Multi-Stage Unsupervised Sequential Deep Learning Method for Ultrasound Strain Elastography
Developed MUSSE-Net, an unsupervised sequential network for displacement and strain estimation, validated on simulation, **in vivo**, and clinical datasets.

- Designed novel architecture including a *Contextual-Aware Feature Encoder*, *TriCrossAttention*, and *Cross-Attentive Sequential Decoder* modules; implemented in PyTorch.

Ongoing Researches

- **Risk-aware Multi-agent Reinforcement Learning** *May 2026 - Present*
  This work focuses on the collaboration of multi-agent systems. How agents collaborate with each other to finish a work effectively.

- **Self-Distillation of Robust VLM** *May 2026 - Present*
  Working on a cross-input self-distillation framework in which a single VLM serves as both teacher and student and stabilises on-policy training.

Work Experience

Adjunct Lecturer, Dept. of CSE, BRAC University. *June 2025 - Present*
Courses taken: Image Processing (CSE428), Digital Electronics (CSE350)

Machine Learning Engineer (Part-time), ACI Ltd. *Apr 2025 - Present*
Projects

- **Medical-ExpertVLM** with explainability and personalized assistance for doctors and medical students.
- Developing a robust OCR and annotation tool using state-of-the art **Qwen2.5VL-VLM** to extract medicine details and bank cheque details from images of a handwritten prescription and cheques.
- Developing an **AI-Assisted** System for **Drug Discovery** and Development

Honors and Awards

- Cockrell Engineering Fellowship (UT Austin) *2026*
- University Merit Scholarship (BUET) *2020, 2021, 2023*
- University Dean's List Scholarship (BUET) *2021, 2022*
- University Stipend (BUET) *2021, 2022*
- **1st** Runner-Up of Undergraduate Project Idea Contest at 25th ICCIT 2022, Bangladesh. [Certificate]
- **57th** in the public leaderboard of DL Sprint - BUET CSE Fest 2024, Bengali AI Math Olympiad a LLM based competition.
- **Best Notebook Award** at the DL Sprint - BUET CSE Fest 2022, Bengali ASR Competition.

Selected Projects

Autonomous Driving with Efficient VLM-based reasoning with LoRA Distilltion Github
The goal of this research is to develop a robust and explainable system for Autonomous Driving (AD) for real time application. I have implemented a Trajectory and path planning for AD with explainability and reasoning. Initially, used **nexar-dashcam AD** dataset and opensource VLM model (Qwen).

Coding Google's PaliGemma VLM from Scratch Github
Implemented Google's opensource ***PaliGemma*** Vision Language Model (VLM) from scratch. Implemented the ***SigLip*** vision encoder, ***KV-cache*** and the ***GemmaLM*** with the multimodal projector. This project was inspired by Umair Jamil's paligemma open-source project.

Autonomous Inventory Robot Github
Developed an autonomous robotic system capable of executing real-time voice commands to identify, retrieve, and transport specified objects. The system integrates ***Google Speech API*** based speech recognition, ***YOLOv5-s*** for real-time object detection with robotic arm manipulation for precise object grasping with a ***Rasphberry-pi*** as the processor. Following successful acquisition, the object is returned to a predefined base location using a Line Following Robot (***LFR***) navigation system.

Deep-Learning-based-Breast-Cancer-Classification-Using-VGGIN Github
Trained the ***VGGIN*** model—a custom deep learning architecture that integrates ***VGG-19*** with the ***Inception*** module, on the BreakHis histopathology dataset. Achieved a test accuracy of 99.628%, demonstrating the model's effectiveness in classifying breast cancer subtypes from histopathological images.

Voice Controlled Wheelchair for Disabled Patients Github
Developed a full scale voice-controlled wheelchair for physically impaired people. ***GMM***-based trained ***VoiceRecognitionV3*** module takes voice commands from the patients in any language and in any accent and moves accordingly. In addition, the wheelchair collision avoidance and emergency help feature.[Video].

Extracting Audio from Muted Video Github
The main goal of this project was to extract the audio signal from a muted video using signal processing methods in MATLAB. The local and global pixel motions were captured using ***Complex Steerable Pyramid*** decomposition. This was originally a project by Abe Davis, MIT [Visual Microphone].

Skills

ML, DL Framework	PyTorch, TensorFlow, Transformers, vLLM, LangChain
Programming	Python, MATLAB, C/C++, Linux, Verilog, Assembly, LaTeX, Git
Simulations and Tools	Raspberry Pi, Arduino, Proteus, Pspice, Quartus, Keil
Frontend Backend	FastAPI, Flusk, Gradio, Android Studio, Kotlin (Basic).